

Exponentials & Polynomials

ANSWER KEY



Laws of exponents

- 1 Simplify $x^5 \cdot x^3$.
 $x^5 \cdot x^3 = x^{5+3} = x^8$.
 - 2 Simplify $\frac{y^9}{y^4}$.
 $\frac{y^9}{y^4} = y^{9-4} = y^5$.
 - 3 Simplify $(2x^3)^4$.
 $(2x^3)^4 = 2^4 x^{12} = 16x^{12}$.
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Negative and fractional exponents

- 4 Simplify x^{-3} , writing your answer without a negative exponent.
 $x^{-3} = \frac{1}{x^3}$.
 - 5 Simplify $16^{3/4}$.
 $16^{3/4} = (16^{1/4})^3 = 2^3 = 8$ (since $16^{1/4} = 2$, because $2^4 = 16$).
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Adding, subtracting, and multiplying polynomials

- 6 Simplify $(3x^2 + 2x - 5) + (x^2 - 4x + 7)$.
Combine like terms: $(3x^2 + x^2) + (2x - 4x) + (-5 + 7) = 4x^2 - 2x + 2$.
 - 7 Expand $(x + 3)(x - 5)$.
 $(x + 3)(x - 5) = x^2 - 5x + 3x - 15 = x^2 - 2x - 15$.
 - 8 Expand $(2x - 1)^2$.
 $(2x - 1)^2 = (2x)^2 - 2(2x)(1) + 1^2 = 4x^2 - 4x + 1$.
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Factoring polynomials

- 9 Factor $x^2 - 16$ completely.
This is a difference of squares: $x^2 - 16 = (x - 4)(x + 4)$.
- 10 Factor $2x^2 + 8x$ completely.
Factor out the greatest common factor: $2x^2 + 8x = 2x(x + 4)$.

- 11 Factor $x^3 - 4x$ completely.

Factor out x : $x^3 - 4x = x(x^2 - 4)$, and then factor the remaining difference of squares: $x(x - 2)(x + 2)$.

Polynomial division and the remainder theorem

- 12 Find the remainder when $f(x) = x^3 - 2x^2 + x - 5$ is divided by $(x - 2)$, using the remainder theorem.
By the remainder theorem, the remainder equals $f(2)$: $f(2) = 8 - 8 + 2 - 5 = -3$.
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Exponential growth vs. polynomial growth

- 13 Two functions are given: $f(x) = x^3$ and $g(x) = 2^x$. Explain which function eventually grows faster as x increases, and estimate the crossover point using the values $f(10) = 1000$, $g(10) = 1024$, $f(8) = 512$, $g(8) = 256$.

Exponential functions eventually always outgrow polynomial functions as $x \rightarrow \infty$, regardless of the polynomial's degree, because the exponential's rate of increase itself keeps growing while the polynomial's relative rate of increase slows. Comparing the given values: at $x = 8$, $f(8) = 512 > g(8) = 256$ (polynomial ahead), but by $x = 10$, $g(10) = 1024 > f(10) = 1000$ (exponential has overtaken it) — so the crossover happens between $x = 8$ and $x = 10$.

Modeling with exponential functions

- 14 An investment of 5000 dollars grows according to $A(t) = 5000(1.06)^t$, where t is the number of years. Find the value of the investment after 5 years, to the nearest dollar, and state the annual growth rate as a percentage.
 $A(5) = 5000(1.06)^5 = 5000(1.3382) \approx 6691$ dollars. The annual growth rate is 6%, since the growth factor is $1.06 = 1 + 0.06$.
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Applying the zero product property to polynomial equations

- 15 Solve $x^3 - 5x^2 + 6x = 0$ by factoring completely and applying the zero product property.
Factor out x : $x(x^2 - 5x + 6) = 0$, then factor the quadratic: $x(x - 2)(x - 3) = 0$. By the zero product property, $x = 0$, $x = 2$, or $x = 3$.
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Multiplying binomials and trinomials

- 16 Expand $(x + 2)(x^2 - 3x + 4)$.

Distribute each term:

$$x(x^2 - 3x + 4) + 2(x^2 - 3x + 4) = x^3 - 3x^2 + 4x + 2x^2 - 6x + 8 = x^3 - x^2 - 2x + 8.$$

- 17 Expand $(2x - 3)(x + 5)$.
 $(2x - 3)(x + 5) = 2x^2 + 10x - 3x - 15 = 2x^2 + 7x - 15$.
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Comparing exponential models

- 18 Two savings accounts grow according to $A(t) = 1000(1.03)^t$ and $B(t) = 1000(1.05)^t$. Explain which account grows faster, and find the difference in value after 10 years, to the nearest dollar.
- Account B grows faster because it has a larger growth factor ($1.05 > 1.03$, meaning a higher annual interest rate). After 10 years: $A(10) = 1000(1.03)^{10} \approx 1344$, $B(10) = 1000(1.05)^{10} \approx 1629$.
Difference: $1629 - 1344 = 285$ dollars.
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Simplifying expressions with multiple exponent rules

- 19 Simplify $\frac{(x^3y^2)^2}{x^4y}$.
Expand the numerator: $(x^3y^2)^2 = x^6y^4$. Divide: $\frac{x^6y^4}{x^4y} = x^{6-4}y^{4-1} = x^2y^3$.
- 20 Simplify $\left(\frac{2x^2}{y}\right)^3$.
 $\left(\frac{2x^2}{y}\right)^3 = \frac{2^3(x^2)^3}{y^3} = \frac{8x^6}{y^3}$.
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Half-life and decay applications

- 21 A radioactive substance has a half-life of 8 years, starting with 200 grams. Write a model for the remaining amount after t years, and find the amount remaining after 24 years.
- Model: $A(t) = 200\left(\frac{1}{2}\right)^{t/8}$. After 24 years: $A(24) = 200\left(\frac{1}{2}\right)^{24/8} = 200\left(\frac{1}{2}\right)^3 = 200\left(\frac{1}{8}\right) = 25$ grams.